

Figure Skating One-Way Anova Worksheet

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Figure skating is a sport where athletes perform choreographed routines on ice to music, combining athletic skill with artistic expression. Skaters compete in different categories like men's singles, women's singles, pairs (a man and a woman skating together), and ice dance (which focuses more on rhythm and movement). Each skater or team performs two routines: the short program and the free skate. The short program is a shorter routine with a list of required moves that every skater must include. Judges rate how well each move is done and how smoothly everything fits with the music. The free skate routine is longer and gives skaters more freedom to show their personal style and creativity. They still need to include certain types of moves, but they can choose how to put them together. Judges give scores for both routines based on how difficult the moves are, how well they're done, and how artistic the performance is. The scores from both routines are added together, and the skater or team with the highest total wins.

Read in the dataset:

```
library(ggplot2)
library(tidyverse)
library(here)
library(dplyr)
```

- 1) Create a box plot with **Year** on the x-axis and **TP** (Total Points) on the y-axis. Group by Year to get a box plot for each year. Describe what you see.

We would like to determine if these means have any statistically significant differences.

- 2) Explain why **One-Way ANOVA** may be a good candidate method to use here.
- 3) Before using ANOVA, we will first need to check for assumptions.

The 3 assumptions are:

- **Independence:** if the samples are collected independently from each other
- **Normality:** if the data is normally distributed (or we have sufficiently large sample sizes such that the Central Limit Theorem holds)
- **Constant Variance:** if the difference between observed and predicted values is equal among data points.

Evaluate the three assumptions. Use your boxplots to assist. You may also want to investigate the data directly (paying close attention to the names of the skaters).

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4. Regardless of your evaluation of the assumptions, we will perform the ANOVA hypothesis test - recognizing that it may not be perfect, but it is still potentially informative.
 - a. State the null hypothesis and alternative hypotheses
 - b. Perform an ANOVA hypothesis test to determine if there are any statistically significant differences in **TP** (Total Points) between the years using the `aov()` function. You will need to change **Year** to a factor.
 - c. Report the F-Statistic, degrees of freedom, and p-value.
 - d. Use the p-value to make a conclusion.

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5. In order to find out what years are statistically discernable from each other we can perform a Post Hoc Tukey Test.

Perform a Post Hoc Tukey test using the `TukeyHSD()` function to determine between which years there is a statistically significant difference in scoring. Report those years. Also report the years that are not significant.

6. The Tukey Test also provides a confidence interval.

Interpret the confidence interval for 2022 and 2014 comparison in the context of this data situation.

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7. Optional: Compact Letter Displays

A **pairwise comparison** is used to test whether differences between group means are statistically significant. In our case, each **year** represents a group. After running a post-hoc test like `TukeyHSD()`, we often summarize the results using **letters**:

- Groups that share a letter have means that are *not* statistically significantly different.
- Groups that do *not* share a letter have means that are statistically significantly different.

This system of labeling is called a **compact letter display**. It helps us quickly interpret which groups are different based on the test results.

Ordering the Groups

Compact letter displays are typically presented in a logical order, most often by ordering groups according to the **sample mean response** in **descending order**.

- Start by filling in the first two columns of this table. Remember to order the groups according to the sample mean Total Points (TP) in descending order.

Year	Mean Total Points (TP)	Group (Letter)
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- To make the rest of the table, we will compare the statistical significance between each of pair of years. If it is not significant, both years gets a letter (following abcd... order). Use a 5% level of significance.

For example, starting with 2022... Any year that is not significant with 2022 gets the letter “a”. Place a letter “a” at 2022 and 2018 since they are not significant with each other. No other year will have an “a” because they are significant with 2022.

Then we move to 2018 with the letter “b” and so on.

- After assigning all letters, compare your results to what you found in the previous question when using `TukeyHSD()`? They should match. If not, double-check your letters!

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- At the 2018 Winter Olympics, many figure skaters began placing their jumps toward the end of their free skate programs to take advantage of the scoring bonus awarded for second-half elements. This strategy led to an overall increase in scores. In response, the International Skating Union (ISU) introduced the “Zagitova Rule,” which limits the number of jumps that can be performed in the second half of a free skate. Additionally, the ISU expanded the Grade of Execution (GOE) scoring range from -3 to +3 to a broader

scale of -5 to +5. Despite these changes, skaters continued to find ways to maximize their scores by attempting more difficult jumps and leveraging the new scoring range to their advantage.

Does the Tukey test you performed support this trend?